The Journal of Open Source Software

tempodisco: an R package for temporal discounting

Isaac Kinley [1¶]

1 Postdoctoral Fellow, Rotman Research Institute, Canada ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Sehrish Kanwal

Reviewers:

- @bkrayfield
- @marberts

Submitted: 03 September 2024
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

Summary

"Temporal discounting" or "delay discounting" refers to the near-universal phenomenon in which humans (and other animals) undervalue future rewards as a function of their delay (Odum, 2011). The degree to which future rewards are discounted in this way is an important individual difference that is related to a range of real-life outcomes (Story et al., 2014). Many methods exist for modeling and quantifying discounting, and `tempodisco` is designed to provide easy access to these.

Statement of need

`tempodisco` is an R package for working with temporal discounting data aimed at behavioural researchers in psychology, economics, and related fields. It is designed to simplify the analysis steps of data cleaning, model fitting, and quantifying discounting. The package implements widely used methods such as computing indifference points from adjusting amount task (Frye et al., 2016), testing for non-systematic discounting per the criteria of Johnson & Bickel (2008), scoring questionnaires according to the methods of Kirby et al. (1999) and Wileyto et al. (2004), and computing model-agnostic measures of discounting such as area under the curve (Myerson et al., 2001) and ED50 (Yoon & Higgins, 2008). The package can also perform approximate Bayesian model selection using a range of discount functions (Franck et al., 2015) for models of both choice-level data and indifference points, and implements more recently proposed methods such as drift diffusion models of discounting (Peters & D'Esposito, 2020) and "fixed-endpoint" psychometric curves (Kinley et al., 2025). Users can also produce simple diagnostic plots to examine model fits. Moreover, the package provides a framework within which users can specify custom discount functions (e.g., discount functions to account for the "magnitude effect"; Green et al. (1997)).

So far, there are relatively few software tools dedicated to working with temporal discounting data: a widely-used Excel-based spreadsheet tool (Kaplan et al., 2016) implements the Kirby et al. (1999) scoring method; a Matlab toolbox (Vincent, 2016) implements hierarchical Bayesian modelling of choice-level data; the R package `discountingtools` (Gilroy, 2024) fits a range of discount functions to indifference point data; and the R package `discAUC` (Friedel, 2024) computes the model-free area under the curve for indifference point data. `tempodisco` provides several important functions not available in these tools, such as modeling choice-level data using a range of discount functions, fitting drift diffusion models, and implementing multiple "choice rules" to link subjective valuations to choice probabilities (Wulff & Bos, 2018). However, `tempodisco` currently fits only individual-level models (vs hierarchical models), only models choices between immediate and delayed rewards (vs choices between two delayed rewards), and only performs maximum-likelihood (vs Bayesian) parameter estimation.

The package is based around a few basic objects such as `td_fn` for discount functions, `td_ipm` for models of indifference points, `td_bclm`/`td_bcnm` for linear/nonlinear models of binary choice data, and `td_ddm` for drift diffusion models. Objects containing models are designed to feel much like R's `lm` and `glm` objects, with methods for generic functions such as `coef`,

`predict`, `residuals`, and `A/BIC`. Thus, users familiar with basic R functionality should find it straightforward to use `tempodisco`. For getting started, the package comes with an example dataset (described in Kinley et al. (2024)) pre-formatted to `tempodisco`'s specifications. With the growing recognition that temporal discounting may be described by a range of functions beyond the standard hyperbolic (Bailey et al., 2021; Franck et al., 2015) and that model fit should be assessed by measures other than R^2 (Gelino et al., 2024), `tempodisco` will allow researchers to easily implement these best practices and will continue to evolve along with best practices in temporal discounting research.

References

Bailey, A. J., Romeu, R. J., & Finn, P. R. (2021). The problems with delay discounting: A critical review of current practices and clinical applications. *Psychological Medicine*, *51*(11), 1799–1806. https://doi.org/10.1017/S0033291721002282

Franck, C. T., Koffarnus, M. N., House, L. L., & Bickel, W. K. (2015). Accurate characterization of delay discounting: A multiple model approach using approximate bayesian model selection and a unified discounting measure. *Journal of the Experimental Analysis of Behavior*, *103*(1), 218–233. https://doi.org/10.1002/jeab.128

Friedel, J. E. (2024). *discAUC: Linear and non-linear AUC for discounting data*. https://CRAN.R-project.org/package=discAUC

Frye, C. C., Galizio, A., Friedel, J. E., DeHart, W. B., & Odum, A. L. (2016). Measuring delay discounting in humans using an adjusting amount task. *Journal of Visualized Experiments: JoVE*, *107*. https://doi.org/10.3791/53584

Gelino, B. W., Strickland, J. C., & Johnson, M. W. (2024). R^2 should not be used to describe behavioral-economic discounting and demand models. *Journal of the Experimental Analysis of Behavior*. https://doi.org/10.1002/jeab.4200

Gilroy, S. (2024). *Discountingtools: Delay discounting tools*. https://doi.org/10.5281/zenodo.7754483

Green, L., Myerson, J., & McFadden, E. (1997). Rate of temporal discounting decreases with amount of reward. *Memory & Cognition*, *25*, 715–723.

Johnson, M. W., & Bickel, W. K. (2008). An algorithm for identifying nonsystematic delay-discounting data. *Experimental and Clinical Psychopharmacology*, *16*(3), 264. https://doi.org/10.1037/1064-1297.16.3.264

Kaplan, B. A., Amlung, M., Reed, D. D., Jarmolowicz, D. P., McKerchar, T. L., & Lemley, S. M. (2016). Automating scoring of delay discounting for the 21-and 27-item monetary choice questionnaires. *The Behavior Analyst*, *39*, 293–304.

Kinley, I., Oluwasola, J., Bagheri, S., & Becker, S. (2024). Goal-directed behaviour is associated with decreased temporal discounting. *PsyArXiv*. https://doi.org/https://doi.org/10.31234/osf.io/qjcwn

Kinley, I., Oluwasola, J., & Becker, S. (2025). Probabilistic models of delay discounting: "Fixed-endpoint" psychometric curves improve plausibility and performance. *Journal of Mathematical Psychology*, *124*, 102902. https://doi.org/https://doi.org/10.1016/j.jmp.2025.102902

Kirby, K. N., Petry, N. M., & Bickel, W. K. (1999). Heroin addicts have higher discount rates for delayed rewards than non-drug-using controls. *Journal of Experimental Psychology: General*, *128*(1), 78. https://doi.org/10.1037//0096-3445.128.1.78

Myerson, J., Green, L., & Warusawitharana, M. (2001). Area under the curve as a measure of discounting. *Journal of the Experimental Analysis of Behavior*, *76*(2), 235–243. https:

DRAFT

//doi.org/10.1901/jeab.2001.76-235

Odum, A. L. (2011). Delay discounting: I'm ak, you're ak. *Journal of the Experimental Analysis of Behavior*, *96*(3), 427–439. https://doi.org/10.1901/jeab.2011.96-423

Peters, J., & D'Esposito, M. (2020). The drift diffusion model as the choice rule in intertemporal and risky choice: A case study in medial orbitofrontal cortex lesion patients and controls. *PLoS Computational Biology*, *16*(4), e1007615. https://doi.org/10.1371/journal.pcbi.1007615

Story, G. W., Vlaev, I., Seymour, B., Darzi, A., & Dolan, R. J. (2014). Does temporal discounting explain unhealthy behavior? A systematic review and reinforcement learning perspective. *Frontiers in Behavioral Neuroscience*, *8*, 76. https://doi.org/10.3389/fnbeh.2014.00076

Vincent, B. T. (2016). Hierarchical bayesian estimation and hypothesis testing for delay discounting tasks. *Behavior Research Methods*, *48*(4), 1608–1620.

Wileyto, E. P., Audrain-McGovern, J., Epstein, L. H., & Lerman, C. (2004). Using logistic regression to estimate delay-discounting functions. *Behavior Research Methods, Instruments, & Computers*, *36*, 41–51. https://doi.org/10.3758/bf03195548

Wulff, D. U., & Bos, W. van den. (2018). Modeling choices in delay discounting. *Psychological Science*, *29*(11), 1890–1894. https://doi.org/10.1177/0956797616664342

Yoon, J. H., & Higgins, S. T. (2008). Turning k on its head: Comments on use of an ED50 in delay discounting research. *Drug and Alcohol Dependence*, *95*(1-2), 169–172. https://doi.org/10.1016/j.drugalcdep.2007.12.011

DRAFT